

List of Publications

Samuel Kaski

September 27, 2026

A Peer-reviewed scientific articles

A1 Journal article, original research

- [1] M. Alakuijala, Y. Gao, G. Ananov, S. Kaski, P. Marttinen, A. Ilin, and H. Valpola. Memento no more: Coaching AI agents to master multiple tasks via hints internalization. *Transactions on Machine Learning Research*, 2026. <https://openreview.net/forum?id=AtCURrC3XA>
- [2] P. Blomstedt, D. Mesquita, O. Rivasplata, J. Lintusaari, T. Sivula, J. Corander, and S. Kaski. Meta-analysis of Bayesian analyses. *Bayesian Analysis*, 21:229–254, 2026.
- [3] Y. Huo, S. P. Dash, R. Stoican, S. Kaski, and M. Sun. Rank-1 approximation of inverse Fisher for natural policy gradients in deep reinforcement learning. *Transactions on Machine Learning Research*, 2026. <https://openreview.net/forum?id=ko8Kn7TS6m>
- [4] J. Karhula, A. Ojanperä, E. Yilmaz, S. Merz, S. Kaski, and R. Salmelin. Machine learning on magnetoencephalography data yields generalizable low-dimensional neural fingerprints that distinguish individuals across task conditions. *Human Brain Mapping*, 47(11):e70618, 2026.
- [5] Z. Rahiminasab, R. Soumi, A. Klami, and S. Kaski. Point-identification of a robust predictor under latent shift with imperfect proxies. *Transactions on Machine Learning Research*, 2026. <https://openreview.net/forum?id=QFJuVreJDC>
- [6] T. da Silva, B. Bazaluk, E. de Souza da Silva, A. Góis, S. Lahlou, D. Heider, S. Kaski, D. Mesquita, and A. H. Ribeiro. Expert-aided causal discovery of ancestral graphs. *Information Sciences*, 756:123816, 2026.
- [7] Y. Yang, A. Rehn, S. Katt, A. Honkela, and S. Kaski. An interactive framework for finding the preferred trade-off in differential privacy. *Transactions on Machine Learning Research*, 2026. <https://openreview.net/forum?id=ire2TQNxfv>
- [8] M. Alakuijala, R. McLean, I. Woungang, N. Farsad, S. Kaski, P. Marttinen, and K. Yuan. Video-language critic: transferable reward functions for language-conditioned robotics. *Transactions on Machine Learning Research*, 2025. <https://openreview.net/pdf?id=jJOVpnNrEp>
- [9] A. Jain, M. Heinonen, H. Käsänen, J. Sipilä, and S. Kaski. Multi-target property prediction and optimization using latent spaces of generative model. *Machine Learning: Science and Technology*, 6:025042, 2025.
- [10] M. A. Masood, A. A. Nagaraja, K. Belaid, N. Mesens, H. Ceulemans, S. Kaski, D. Herman, and M. Heinonen. VitroBert: modeling DILI by pretraining BERT on in vitro data. *Journal of Chemoinformatics*, 17:119, 2025.
- [11] A. Aushev, T. Tran, H. Pesonen, A. Howes, and S. Kaski. Likelihood-free inference in state-space models with unknown dynamics. *Statistics and Computing*, 34:27, 2024.
- [12] L. Filstroff, I. Sundin, P. Mikkola, A. Tiulpin, J. Kylmäoja, and S. Kaski. Targeted active learning for Bayesian decision-making. *Transactions on Machine Learning Research*, 2024. <https://openreview.net/forum?id=KxPjuiMgmm>
- [13] J. Haakana, S. Merz, S. Kaski, H. Renvall, and R. Salmelin. Bayesian reduced rank regression models generalisable neural fingerprints that differentiate between individuals in magnetoencephalography data. *European Journal of Neuroscience*, 59:2320–2335, 2024.

- [14] I. H. Haraldsen, C. Hatlestad-Hall, C. Marra, H. Renvall, F. Maestu, J. Acosta-Hernandez, S. Alfonsin, V. Andersson, A. Anand, V. Ayllon, A. Babic, A. Belhadi, C. Birk, R. Bruna, N. Caraglia, C. Carrarini, E. Christensen, A. Cicchetti, S. Daugbjerg, R. Di Bidino, A. Diaz-Ponce, A. Drews, G. M. Giuffre, J. Georges, P. Gil-Gregorio, D. Gove, T. M. Govers, H. Hallock, M. Hietanen, L. Holmen, J. Hotta, S. Kaski, R. Khadka, A. S. Kinnunen, A. M. Koivisto, S. Kulashkekhar, D. Larsen, M. Liljestrom, P. G. Lind, A. Marcos Dolado, S. Marshall, S. Merz, F. Miraglia, J. Montonen, V. Mantynen, A. R. Oksengard, J. Olazaran, T. Paajanen, J. M. Pena, L. Pena, D. Irabien Peniche, A. S. Perez, M. Radwan, F. Ramirez-Torano, A. Rodriguez-Pedrero, T. Saarinen, M. Salas-Carrillo, R. Salmelin, S. Sousa, A. Suyuthi, M. Toft, P. Toharia, T. Tveitstol, M. Tveter, R. Upreti, R. J. Vermeulen, F. Vecchio, A. Yazidi, and P. M. Rossini. Intelligent digital tools for screening of brain connectivity and dementia risk estimation in people affected by mild cognitive impairment: the ai-mind clinical study protocol. *Frontiers in Neurorobotics*, 17, 2024. doi:10.3389/fnbot.2023.1289406
- [15] A. Klami, T. Damoulas, O. Engkvist, P. Rinke, and S. Kaski. Virtual laboratories: Transforming research with AI. *Data-Centric Engineering*, 5:e19, 2024.
- [16] J. Menke, Y. Nahal, E. J. Bjerrum, M. Kabeshov, S. Kaski, and O. Engkvist. Metis - a python-based user interface to collect expert feedback for generative chemistry models. *Journal of Chemoinformatics*, 16:100, 2023.
- [17] P. Mikkola, O. A. Martin, S. Chandramouli, M. Hartmann, O. A. Pla, O. Thomas, H. Pesonen, J. Corander, A. Vehtari, S. Kaski, P.-C. Bürkner, and A. Klami. Prior knowledge elicitation: The past, present, and future. *Bayesian Analysis*, 19:1129 – 1161, 2024.
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- [19] L. Prediger, J. Jälkö, A. Honkela, and S. Kaski. Collaborative learning from distributed data with differentially private synthetic data. *BMC Medical Informatics and Decision Making*, 24:167, 2024.
- [20] Z. R. Yousefi, T. Vuong, M. Alghossein, T. Ruotsalo, G. Jacucci, and S. Kaski. Entity footprinting: Modeling contextual user states via digital activity monitoring. *ACM Transactions on Interactive Intelligent Systems*, 14:9, 2024.
- [21] A. Aushev, A. Putkonen, G. Clarté, S. Chandramouli, L. Acerbi, S. Kaski, and A. Howes. Online simulator-based experimental design for cognitive model selection. *Computational Brain & Behavior*, 6:719–737, 2023.
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- [29] Z. Yang, Y. Chen, D. Sedov, S. Kaski, and J. Corander. Stochastic cluster embedding. *Statistics and Computing*, 33:12, 2023.
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- [32] T. Cui, A. Havulinna, P. Marttinen, and S. Kaski. Informative Bayesian neural network priors for weak signals. *Bayesian Analysis*, 17:1121–1151, 2022.
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